

NT5

Дано: $\xi \sim R[0, 20]$; $\bar{x}_n$ - выборка

$$M\xi = \frac{0+20}{2} = \frac{3}{2} \cdot 0$$

$$p(x) = \frac{1}{20-0} = \frac{1}{20}$$

$$D\xi = \frac{(20-0)^2}{12} = \frac{20^2}{12}$$

а) ОММ:

$$M\xi(\theta) = \bar{x} \Rightarrow \hat{\theta}_1 = \frac{2}{3} \bar{x}$$

ОМП:

$$L(\theta) = \prod_{i=1}^n p(x_i, \theta) = \prod_{i=1}^n \frac{1}{20} = \frac{1}{20^n}$$

$$x_{\min} \geq 0 \text{ и } x_{\max} \leq 20$$

$$\Rightarrow \frac{x_{\max}}{2} \leq x_{\min} \text{ , т.к. } L(\theta) \text{ - убывающ.}$$

$$\sup L(\theta) = \sup L\left(\frac{x_{\max}}{2}\right) \Rightarrow \hat{\theta}_2 = \frac{x_{\max}}{2}$$

б) Проверка на несмещённость:

$$M\hat{\theta}_1 = M\left(\frac{2}{3} \cdot \frac{1}{n} \sum_{i=1}^n x_i\right) = \frac{2}{3n} \cdot n \cdot M\xi = \frac{2}{3} \cdot \frac{3}{2} \cdot 0 = 0$$

- явл. несмещён.

$$= \frac{1}{4} \int_0^{\theta} (t+\theta)^2 \frac{n}{\theta^n} t^{n-1} dt = \frac{n}{\theta^n} \cdot \frac{1}{4} \int_0^{\theta} (t^2 + 2\theta t + \theta^2) t^{n-1} dt$$

$$= \frac{1}{4} \cdot \frac{n}{\theta^n} \cdot \left(\frac{\theta^{2+n}}{2+n} + \frac{2\theta^{2+n}}{n+1} + \frac{\theta^{2+n}}{n} \right) =$$

$$= \frac{1}{4} \theta^2 \cdot \frac{n(n^2 + n + 2n^2 + 4n + n^2 + 2n + n + 2)}{n(n+1)(n+2)} =$$

$$= \frac{1}{4} \theta^2 \frac{4n^2 + 8n + 2}{(n+1)(n+2)} = \frac{1}{2} \theta^2 \frac{2n^2 + 4n + 1}{(n+1)(n+2)}$$

$$\mathcal{D} \tilde{\theta}_2^2 = \frac{1}{2} \theta^2 \frac{2n^2 + 4n + 1}{(n+1)(n+2)} - \theta^2 \frac{4n^2 + 4n + 1}{4(n+1)^2} =$$

$$= \theta^2 \left(\frac{4n^3 + 8n^2 + 2n + 4n^3 + 8n + 2 - 4n^3 - 4n^2 -}{4(n+1)^2(n+2)} \right) =$$

$$= \theta^2 \frac{n}{4(n+1)^2(n+2)} \xrightarrow{n \rightarrow +\infty} 0 \quad - \text{составительная}$$

$$\text{Т.к. } \tilde{\theta}_2^1 = \frac{2n+2}{2n+1} \cdot \tilde{\theta}_2 \quad \vee \quad \left(\frac{2n+2}{2n+1} \right)^2 \xrightarrow{n \rightarrow +\infty} 1,$$

$$\text{то } \mathcal{D} \tilde{\theta}_2^1 \xrightarrow{n \rightarrow +\infty} 0.$$

$$c) D\tilde{\theta}_2' = \left(\frac{2n+2}{2n+1}\right)^2 \cdot \theta^2 \cdot \frac{n}{4(n+1)^2(n+2)} = \theta^2 \cdot \frac{n}{(2n+1)^2(n+2)}$$

$$D\tilde{\theta}_1 = \frac{4}{9n} \cdot \frac{\theta^2}{12} = \frac{\theta^2}{27n}$$

$$27n^2 < (4n^2 + 4n + 1)(n+2)$$

$$27n^2 < 4n^3 + 8n^2 + 4n^2 + n + 8n + 2$$

$$0 < 4n^3 - 15n^2 + 9n + 2 \quad \text{при } n \geq 3$$

$$\Rightarrow \frac{1}{27n} > \frac{n}{(2n+1)^2(n+2)} \Rightarrow D\tilde{\theta}_1 > D\tilde{\theta}_2'$$

при $n \geq 3$

$\Rightarrow$ Оценка $\tilde{\theta}_2'$ более эффективна.

$$d) \xi \sim R[0, 2\theta]$$

$$\eta = \frac{\xi}{\theta} \sim R[1, 2], \quad y_i = \frac{x_i}{\theta}$$

$$f(\vec{x}_n, \theta) = \frac{x_{\max}}{\theta} = y_{\max} \sim P(t) = F_R^n(t)$$

$$P(t) = F^n(t) = (t-1)^n$$

$$\varphi(t) = n(t-1)^{n-1}, \quad f(\vec{x}_n, \theta) = \frac{x_{\max}}{\theta} \sim \varphi(t)$$

$$t_1 = q_{\frac{1-\beta}{2}} \quad t_2 = q_{\frac{1+\beta}{2}} \quad t_1$$

$$P\left(t_1 < \frac{x_{\max}}{\theta} < t_2\right) = \beta \int_1^{\frac{x_{\max}}{\theta}} \varphi(t) dt = 0,025$$

$$P\left(\frac{x_{\max}}{t_2} < \theta < \frac{x_{\max}}{t_1}\right) = \beta$$

$$t_1 = 1 + (0,025)^{1/n}$$

$$t_2 = 1 + (0,975)^{1/n}$$

Итог: точный довер. интервал: $\left(\frac{x_{\max}}{t_2}, \frac{x_{\max}}{t_1}\right)$

е) Используем $\tilde{\theta}_1$.

Условия простейшей ЦПТ выполнены
 $\Rightarrow$ работает сб-бо

$$\frac{\tilde{x}_k - \alpha_k}{\sqrt{\alpha_k - \alpha_k^2}} \sqrt{n} \rightsquigarrow N(0,1)$$

$$\text{или для } \tilde{\theta}_1 = g(\tilde{x}) = \frac{2}{3} \tilde{x}_1$$

$$\sqrt{n} \frac{g(\tilde{x}) - g(\alpha)}{\sigma(\tilde{x})} \rightsquigarrow N(0,1), \text{ где}$$

$$\sigma^2(\alpha) = \nabla^T g(\alpha) K \nabla g(\alpha)$$

$$\sigma^2(\alpha) = \frac{2}{3} \cdot (\alpha_2 - \alpha_1^2) \cdot \frac{2}{3} = \frac{4}{9} (\alpha_2 - \alpha_1^2)$$

$$\sqrt{n} \frac{3(\tilde{\theta}_1 - \theta)}{2 \sqrt{\tilde{x}_1 - \tilde{x}_1^2}} \rightsquigarrow N(0,1)$$

$$t_1 = U_{\frac{1-\beta}{2}}, \quad t_2 = U_{\frac{1+\beta}{2}}$$

$$t_1 < \sqrt{n} \frac{3(\tilde{Q}_1 - Q)}{2 \sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}} < t_2$$

$$\left(\tilde{Q}_1 - \frac{2}{3} t_2 \frac{\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}}{\sqrt{n}}, \tilde{Q}_1 - \frac{2}{3} t_1 \frac{\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}}{\sqrt{n}} \right)$$

f, g, h) - see pair "f-g-h-points.ipynb"

$$M\tilde{\theta}_2 \approx M\left(\frac{x_{\max}}{2}\right) = \frac{1}{2} M x_{\max}$$

$$P(x_{\max} < y) = P(y) = F^n(y)$$

$$p(y) = n F^{n-1}(y) \cdot p(y) = \frac{n}{\theta} \cdot \left(\frac{y-\theta}{\theta}\right)^{n-1}$$

$$M\tilde{\theta}_2 = \frac{1}{2} \int_{-\infty}^{+\infty} y \cdot \frac{n}{\theta^n} (y-\theta)^{n-1} dy = \frac{1}{2} \frac{n}{\theta^n} \int_{\theta}^{2\theta} y (y-\theta)^{n-1} dy$$

$$= \left\{ \begin{array}{l} y - \theta = t \end{array} \right\} = \frac{1}{2} \frac{n}{\theta^n} \int_0^{\theta} (t+\theta) t^{n-1} dt =$$

$$= \frac{1}{2} \frac{n}{\theta^n} \left(\frac{\theta^{n+1}}{n+1} + \frac{\theta^{n+1}}{n} \right) = \theta \frac{2n+1}{2(n+1)}$$

— не явл. несмещ.

$$\Rightarrow \tilde{\theta}_2 = \frac{2n+2}{2n+1} \cdot \frac{x_{\max}}{2}$$

Проверка на состоятельность:

$$D\tilde{\theta}_1 = \left(\frac{2}{3}\right)^2 \cdot \frac{1}{n^2} \cdot n D\xi = \frac{4}{9n} \cdot \frac{\theta^2}{12} \xrightarrow{n \rightarrow +\infty} 0$$

состоятельность на

$$M\tilde{\theta}_2^2 = \frac{1}{4} M x_{\max}^2$$

$$M\tilde{\theta}_2^2 = \frac{1}{4} \int_{\theta}^{2\theta} y^2 \frac{n}{\theta^n} (y-\theta)^{n-1} dy =$$